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The chemisorption of hydrogen on the (111) and (110)-(1×2) surfaces of iridium and platinum

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The chemisorption of hydrogen on both the Ir(111) and Pt(110)-(1×2) surfaces has been examined under ultrahigh vacuum conditions with thermal desorption mass spectrometry, LEED, and contact potential difference measurements. No ordered adsorbate superstructures were observed on either surface at any fractional coverage and at surface temperatures from 100 to 700 K, and the (1×2) reconstruction of the Pt(110) surface was stable in all cases. Hydrogen adsorbs dissociatively on the Ir(111) surface, the adsorption reaction described by second-order Langmuir kinetics with an initial probability of adsorption of 7×10^{-3}. The rate parameters describing the second-order desorption reaction of hydrogen from the Ir(111) surface are weakly dependent on coverage between fractional coverages of 0.1 and 0.3, and are given by $E_d \simeq 12.6$ kcal mol^{-1} and $k_d^{(2)} \simeq 2\times10^{-6}$ cm^2 s^{-1}. Beyond a fractional coverage of 0.3, however, both rate parameters decrease with increasing coverage. Hydrogen adsorbs dissociatively on the Pt(110)-(1×2) surface into two distinct β_2 and β_1 adstates, and the ratio of the saturation densities of these two states, β_2:β_1, is 1:2. Adsorption into the higher binding energy β_2 adstate is described by first-order Langmuir kinetics with an initial probability of adsorption of 0.46, whereas adsorption into the β_1 adstate is described by second-order Langmuir kinetics with an "initial" probability of adsorption of 0.022. The rate parameters describing the desorption reaction of hydrogen from the Pt(110)-(1×2) surface are *strongly* dependent on the coverage. In the coverage regime characteristic of the β_2 adstate ($\theta \leqslant 0.32$) the rate parameters are approximately symmetric about one-half of saturation of this state. Specifically, from the values for the zero-coverage limit of $E_d \simeq 18$ kcal mol^{-1} and $k_d^{(2)} \simeq 10^{-4}$ cm^2 s^{-1}, the parameters first increase to maximum values of $E_d \simeq 26.5$ kcal mol^{-1} and $k_d^{(2)} \simeq 0.3$ cm^2 s^{-1} at $\theta = 0.15$, and subsequently decrease approximately to the values for the zero-coverage limit at $\theta = 0.32$ In the coverage regime characteristic of the β_1 adstate ($\theta > 0.32$), the activation energy decreases continuously with increasing coverage from a value of $E_d \simeq 17$ kcal mol^{-1} at $\theta = 0.35$, whereas the preexponential factor remains essentially constant with a value of 3×10^{-4} cm^2 s^{-1}. The contact potential difference for hydrogen on Pt(110)-(1×2) increases continuously with coverage to a value of 0.17 eV at $\theta = 0.30$. As the coverage increases further, however, it decreases continuously approaching a value of -0.50 eV at saturation. Probable binding states for the β_2 and β_1 adstates on the Pt(110)-(1×2) surface are inferred from both the adsorption and desorption kinetics and the contact potential difference measurements. Comparisons of the results obtained on the (111) and (110)-(1×2) surfaces of both iridium and platinum suggest strongly that *local* surface structure (e.g., "step" sites vs terrace sites) has a profound influence on the kinetics of adsorption of hydrogen on these surfaces. Surface structure apparently also has a profound influence on the desorption kinetics of hydrogen via the mediation of adatom–adatom interactions. Whereas both attractive and repulsive interactions are clearly manifest within the β_2 adstates on the (110)-(1×2) surfaces, only repulsive interactions are apparent on the (111) surfaces and for the β_1 adstates on the (110)-(1×2) surfaces.

I. INTRODUCTION

The study of the interaction of hydrogen with transition metal surfaces is of considerable fundamental importance. For example, hydrogen plays a central role in a multitude of technologically important catalytic reactions that occur on transition metal surfaces, such as hydrocarbon processing and Fischer–Tropsch synthesis. Other topics of technological interest include the embrittlement of materials by hydrogen and the possible use of transition metals as hydrogen storage media. Furthermore, the most reliable theoretical calculations can be carried out to describe these systems. Not surprisingly, a multitude of experimental investigations have been concerned with the chemisorption of hydrogen on various single crystalline surfaces of transition metals,[1] including the Group VIII metals.[2–22]

The primary thrust of the work described here is con-

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cerned with a detailed examination of the effects of surface structure on the dynamics of the elementary surface reactions that describe the adsorption and desorption of hydrogen. Previous studies have indicated that the presence of surface defects in the form of atomic steps can influence profoundly the kinetics of the adsorption and desorption reactions. For example, the initial probability of adsorption at step sites is nearly unity at surface temperatures of approximately 100 K and gas temperatures of approximately 300 K, whereas the corresponding probability on the "perfect" (111) surfaces of fcc transition metals is typically lower by an order of magnitude.[2–17] Recent work on the Ni(111) surface and the considerably more corrugated Ni(110) surface indicates that the low probability of adsorption on the (111) surface can be explained by the presence of an activation barrier to dissociative adsorption, whereas adsorption on the (110) surface is essentially unactivated.[16] The activation barrier on the Ni(111) surface is on the order of 2 kcal mol^{-1}, as judged by a continuous increase of the probability of adsorption with increasing normal translational energy of the H_2. This viewpoint is supported by work examining the adsorption kinetics of hydrogen on the stepped Ni(997) surface.[17] The effect of surface stucture of the kinetics of the desorption reaction is manifest in a higher activation energy of desorption in the low coverage limit for hydrogen adsorbed at step sites, compared to hydrogen adsorbed at (111) terrace sites.[3,9,13,15] Clearly, both the binding energy of the hydrogen adatom and the activation barrier of dissociative adsorption are influenced by the surface structure.

We present here new experimental data concerning the chemisorption of hydrogen on both the Ir (111) and the Pt(110)-(1×2) surfaces. These data are compared and contrasted both with data obtained previously in our laboratory concerning the chemisorption of hydrogen on the Ir(110)-(1×2) surface,[18] and with data obtained by a number of investigators concering the chemisorption of hydrogen on the Pt(111) surface.[2–9,19–22] The comparison of the (111) and (110)-(1×2) surfaces of both iridium and platinum is motivated in part by similarities (in addition to the obvious differences) in the *local* surface structure between the two surfaces. In particular, the clean (110) surfaces of both iridium[23] and platinum[24] reconstruct to form (111) microfacets which are three atomic rows in width, and which are inclined at an angle of 109.5° with respect to one another (cf., Fig. 1). Consequently, the presence of structurally similar threefold hollow adsites on each surface might be manifest in similar adsorption and desorption kinetics. However, the corrugated (110)-(1×2) surface possesses unique adsites as a consequence of its geometry (e.g., high coordination "trough" sites, cf., site B of Fig. 1) which might influence greatly the rates of both the adsorption and desorption of hydrogen, especially at low surface coverages. For example, it has been observed that hydrogen prefers to reside in the highest coordination adsites that are available on a surface.[1]

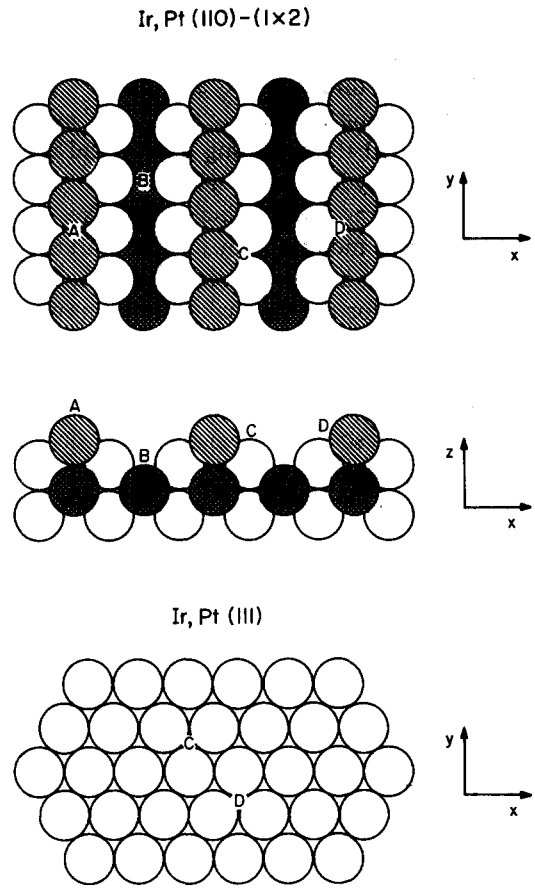


FIG. 1. Structural models for the (110)-(1×2) and (111) surfaces of iridium and platinum. A and B represent two possible locations for the hydrogen adatoms in the higher binding energy β_2 adstates observed on the (110)-(1×2) surfaces (see the text), A being a twofold bridge site, and B being a pseudo-fourfold hollow site, the latter being the more probable location. C and D denote the preferred hydrogen binding sites both on the (111) surfaces and for the β_1 adstates on the (110)-(1×2) surfaces which, on both surfaces, are inequivalent threefold hollow sites.

II. EXPERIMENTAL PROCEDURES

The experiments described here were performed in an ion pumped, stainless steel bell jar that has been described in detail previously.[25] The base pressure of the bell jar is below 5×10^{-11} Torr of reactive gases. Facilities are available for experimental studies involving mass spectrometry, low-energy electron diffraction (LEED), contact potential difference measurements, x-ray and UV-photoelectron spectoscopies, and Auger electron spectroscopy. The Ir(111) and Pt(110) crystals were cut from single crystalline boules and were polished to within 0.5° of the appropriate orientations using standard metallographic techniques. The iridium (platinum) crystal was cleaned *in situ* by argon ion sputtering, heating in 5×10^{-7} Torr of oxygen at a temperature of 1100 K (700 K), and annealing to 1600 K (1400 K). Surface cleanliness was verified by Auger electron and x-ray photoelectron spectroscopies. Special care was taken to reduce the concentration of the silicon impurity in the platinum sample to a negligible level, since Niehus and Comsa[26] have shown that its presence is related to the formation of a "subsurface oxide" on Pt(111).

Research grade (99.995%) hydrogen and deuterium were utilized without further purification. Deuterium was employed in a number of desorption experiments in order to increase the sensitivity (i.e., the signal-to-background ratio) of the mass spectrometric detection and to examine qualita-

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tively the isotopic mixing of the adatoms on the surface. Exposures were effected either by backfilling the chamber or by employing a (calibrated) directional beam doser.[27] The beam doser consists of a multichannel array of capillaries, and the gases are admitted to the dosing line behind this array via a separate capillary leak from a glass storage bulb. The gas pressure in the storage bulb (which determines the pressure at the doser) was monitored by a MKS Baratron gauge. A glass enclosure was installed around the ion source of the mass spectrometer for the thermal desorption mass spectrometric (TDMS) measurements.[28] This glass enclosure permits the desorption products to be sampled preferentially from the front of the crystal, thus minimizing spurious signals originating from desorption from the edges of the crystal, the mounting wire leads, and the manipulator.

The single crystals were spotwelded to tantalum wire leads which were mounted on a precision manipulator. Heating was effected by passing current through the wire leads/crystal circuit, while cooling was accomplished by conduction through the wire leads to support blocks that were in thermal contact with a reservoir of flowing liquid nitrogen. The temperature was measured by a W/5%Re–W/26%Re thermocouple spotwelded to the back of the crystal and referenced to an icepoint junction. Accurate temperature control was maintained by the use of an optimal control scheme, described elsewhere,[29] which utilizes a laboratory microcomputer as the control element. This control scheme is well suited for the generation of linear temperature ramps that were used here in connection with the measurement of thermal desorption spectra. A wide range of heating rates can be realized, and the temporal temperature profiles are sufficiently reproducible to permit the averaging of spectra (following suitable baseline corrections) in order to increase the signal-to-noise ratio. The temperature control software has been designed both to control and to monitor the mass spectrometer, and single routine provides data acquisition as well as contol of the experiment.[30]

Contact potential difference (CPD) measurements were made by employing the ac retarding potential technique of Nathan and Hopkins.[31] A high-stability, low-noise, and low-drift power supply (Kepco PCX 100-0.2 MAT) was utilized to provide the appropriate potentials for the (LEED) electron gun (Varian 981-2125). A standard carbon battery provided the retarding bias, and a biquad bandpass filter (gain of 10, Q factor of 100, tuned to 7 kHz) was used to condition the output from the collector (i.e., the crystal). When tuned properly, noise margins were 5 mV, and the CPD measurements were reproducible to within less than 10 mV at a bandwidth of 10 Hz (beam energy $\sim$45 V and beam current $\sim 0.1\ \mu$A).

III. RESULTS

A. Hydrogen on Ir(111)

1. LEED observations

Under all conditions examined, i.e., at any fractional coverage and at surface temperatures as low as 100 K, no ordered adsorbate superstructures were observed, as judged by the absence of any fractional order LEED beams. This result is not surprising since ordered superstructures of hydrogen on hexagonally close-packed surfaces are rare [excluding the degenerate (1×1) overlayer, the presence of which has been inferred for hydrogen on Ru(001),[32] Pd(111),[13] and Pt(111)[33]]. A notable exception is the $p(2\times2)$-2H superstructure that has been observed for hydrogen on the Ni(111) surface.[34,35] Based solely on our (qualitative) LEED observations, we cannot rule out the possiblity of an ordered (1×1) overlayer structure forming on the Ir(111) surface.

2. Desorption kinetics

Thermal desorption spectra of H_2 from Ir(111), corresponding to ten different initial coverages that were obtained by varying the hydrogen exposure at 100 K, are shown in Fig. 2. The average heating rate β for each desorption spectrum was 20 K s^{-1}.[37] Several qualitative features of the desorption kinetics are immediately apparent from Fig. 2. Consistent with the availability of only one high coordination adsite,[38] the threefold hollow site, the desorption spectra exhibit one desorption feature or "state." The peak temperature is observed to decrease continuously with increasing coverage, which is consistent with a second-order (adatom recombination) desorption reaction.[39] Broadening of the desorption spectra (i.e., the FWHM increases more than the maximum intensity) is clearly evident at the higher coverages corresponding to exposures above approximately 10–20 L (1L = 1 Langmuir $\equiv 10^{-6}$ Torr s), indicative of the presence of repulsive adatom–adatom interactions. These desorption spectra are qualitatively similar to those obtained on other hexagonally close-packed surfaces of the Group VIII metals, e.g., Pt(111),[19] Rh(111),[40] and Ru(001).[41,42] However, the Ni(111) surface provides an exception, the desorption spectra from this surface"splitting" into two "states" for fractional coverages above one-half of a monolayer [i.e., the coverage corresponding to an optimally ordered (2×2) superstructure].[35] Presumably, on Ni(111),

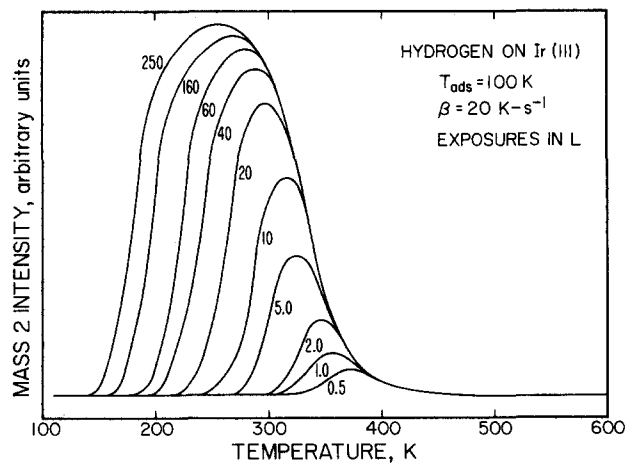


FIG. 2. Thermal desorption spectra of hydrogen from the Ir(111) surface parametric in exposure of hydrogen. The surface temperature during adsorption was 100 K, and the average heating rate (β) for each spectrum was 20 K s^{-1} Ref. 37.

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the splitting of the desorption spectra and the formation of the submonolayer (2×2) superstructure are correlated.

The desorption kinetics were characterized quantitatively by employing an integral method involving variable heating rates.[43] This method permits the assessment of the coverage dependence of the integral kinetic rate parameters, i.e., the activation energy of desorption $[E_d(\theta)]$ and the (second-order) preexponential factor $[k_d^{(2)}(\theta)]$ of the desorption rate coefficient. The variation of the (linear) heating rate permits an evaluation of the rate of desorption (proportional to the mass spectrometric intensity for a chamber that is pumped sufficiently rapidly[39]) as a function of temperature at a *fixed coverage*, the latter of which is determined by integrating the mass spectrometric intensity vs time. Depending on the energetics of the desorption reaction and the reproducibility of the temporal temperature profiles (which permits the averaging of spectra), varying the heating rate by 1–2 orders of magnitude is sufficient for the construction of a set of Arrhenius plots. The activation energy of the desorption reaction for a particular adsorbate coverage can be evaluated from the Arrhenius plot of the rate of desorption as a function of reciprocal surface temperature. By assuming an implicit coverage dependence (i.e., order) of the desorption reaction, the preexponential factor of the desorption rate coefficient can also be evaluated as a function of adsorbate coverage. The major underlying assumption in this analysis is that the desorption reaction procedes via a direct, one-step process, with the rate of this reaction specified completely (and independently) by the surface temperature and the adsorbate concentration.[44]

Arrhenius plots constructed from thermal desorption spectra of D_2 from Ir(111) that embodied two different fixed initial coverages and six different heating rates are shown in Fig. 3. Deuterium, which produces identical desorption spectra to those shown for hydrogen in Fig. 2, was employed to increase the sensitivity of the mass spectrometric detection (especially important at low heating rates). The heating rates were varied from 1.5 to 60 K s^{-1}. Two different initial fractional coverages, $\theta_0 = 0.15$ and $\theta_0 = 0.47$, were studied in order to emphasize two different coverage regimes. (An optimal signal-to-noise ratio is obtained by confining the analysis to the central part of the desorption peak, i.e., $0.2\theta_0 \leqslant \theta \leqslant 0.8\theta_0$.) The former emphasized the low coverage limit, whereas the latter emphasized the low to moderate coverage regime. Higher coverages were not examined since contamination of the surface by background gases (primarily CO) became significant. This was due to a combination of the relatively high exposures necessary to obtain a high coverage (exposures ≥ 100 L) and the low heating rates (total time for a complete spectrum ≥ 200 s) required by this method of analysis.

Making use of the data displayed in Fig. 3, the activation energy of desorption and the preexponential factor of the desorption rate coefficient have been evaluated as a function of coverage. These results are shown in Fig. 4. The preexponential factor was evaluated by assuming that the saturation density of hydrogen is 1.57×10^{15} atoms cm^{-2} (the uncertainty associated with this assumption is negligible compared to other experimental uncertainties; see Sec. III A 3), and that the desorption reaction is second order in the fractional coverage of hydrogen adatoms.

The coverage dependence of the rate parameters is evident from an inspection of Fig. 4. For a fractional coverage below approximately 0.10, a significant variation in the rate parameters is observed—the activation energy of desorption decreases from 18 to 14 kcal mol^{-1}, whereas the preexponential factor decreases from 1×10^{-3} to 1×10^{-5} cm^2 s^{-1}. Between fractional coverages of approximately 0.1 and 0.3 the rate parameters are essentially independent of coverage and are given by $E_d \simeq 12.6$ kcal mol^{-1} and $k_d^{(2)} \simeq 2\times10^{-6}$ cm^2 s^{-1}. As the fractional coverage increases above 0.3, the rate parameters begin to decrease further. For coverages below approximately 0.1 and above approximately 0.3, a compensation effect is observed with the activation energy and preexponential factor varying in sympathy with coverage.

The strong coverage dependence of the rate parameters

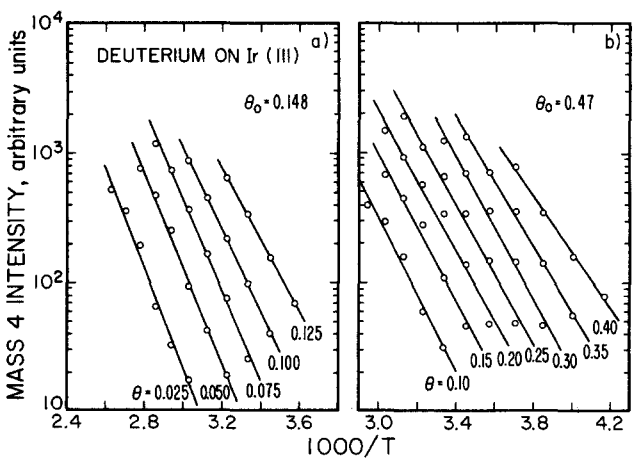


FIG. 3. Arrhenius plots obtained for deuterium desorption from the Ir(111) surface by varying the heating rate and maintaining a constant initial coverage of (a), 0.15; and (b), 0.47.

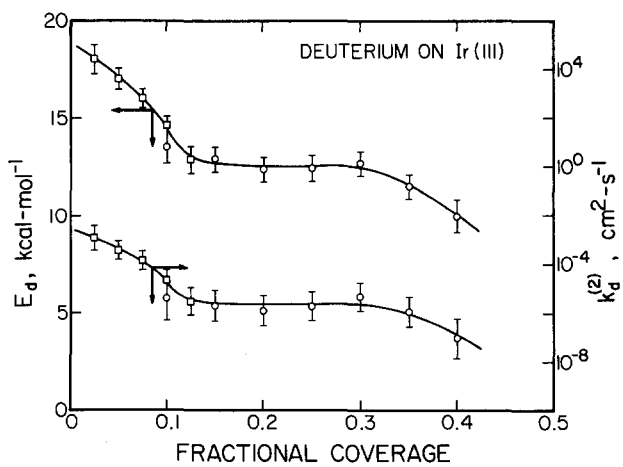


FIG. 4. Activation energy (E_d) and second-order preexpontential factor ($k_d^{(2)}$) of the desorption rate coefficient of D_2 desorption from Ir(111) as a function of fractional surface coverage. The squares were obtained from the data presented in Fig. 3(a) ($\theta_0 = 0.15$), the circles from Fig. 3(b) ($\theta_0 = 0.47$).

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in the limit of low coverages is suggestive of the presence of a small concentration of defect sites at which the binding energy of the hydrogen adatoms is *intrinsically* greater. Since the probability of adsorption at defect sites is significantly greater than that at perfect (111) sites, these defect sites will be populated preferentially at low coverages. Moreover, since hydrogen adatoms are bound more strongly at defect sites, diffusion between adsites will not act to remove this preferential population as molecules are desorbed in a TDMS experiment, i.e., diffusion will act to populate defect sites as opposed to depopulating them. Thus, if defect sites are present, they will dominate the desoption kinetecs at low coverages. This has been observed by Christmann and Ertl[3,45] who examined the chemisorption of hydrogen on Pt(111) and on a stepped Pt[9(111)×(111)] surface (i.e., "defect site" density of 10%). Between fractional coverages of 0.2 and 0.5, both surfaces of platinum exhibited similar activation energies of desorption. However, at coverages below 0.2 the activation energy of desorption on the (111) surface remained virtually constant, whereas on the stepped 9(111)×(111) surface it increased approximately linearly to a value ~ 3 kcal mol^{-1} greater than that observed on the (111) surface in the limit of zero coverage. Since the divergence in the kinetic parameters occurred at $\theta = 0.2$ (i.e., *not* 0.1), it is probable that there was significant mixing between (i.e., recombination of) hydrogen adatoms at step sites and those at (111) terrace sites during the molecular desorption. This behavior of the kinetic parameters observed on the Pt[9(111)×(111)] surface is essentially identical to that observed here on the Ir(111) surface and suggests the presence of a small percentage of defect sites on this Ir(111) surface. Scaling our results to those of Christmann and Ertl[3] would suggest that the Ir(111) surface has a defect density on the order of $\sim 5\%$.

The decrease in the activation energy of desorption at coverages above approximately 0.3 is explained best by the occurrence of repulsive adatom–adatom interactions (presumably "through-metal") at these adsorbate concentrations. These interactions act to decrease the (average) binding energy of the hydrogen adatoms and, hence, the activation energy of desorption. The fact that these interactions appear to become significant at a fractional coverage of approximately 0.3 is of particular interest since the heat of adsorption of hydrogen on the Pt(111) surface is also essentially constant below this coverage.[22] However, as the coverage increases beyond 0.3, the heat of adsorption decreases linearly. These results, for hydrogen on both Ir(111) and Pt(111) suggest that next-nearest-neighbor interactions between hydrogen adatoms are relatively weak, whereas nearest neighbor interactions are repulsive on these surfaces. One can populate a hexagonally close-packed lattice (considering only one of the two inequivalent threefold hollow sites as an adsite[46]) at a fractional coverage of one-third such that no nearest neighbor adatom pairs exist. However, for fractional coverages above one-third, the adlayer must contain nearest neighbor pairs, and the occurrence of repulsive interactions between nearest neighbor hydrogen adatoms acts to decrease the (average) binding energy and, consequently, the activation energy of desorption.

The observation of a $(\sqrt{3}\times\sqrt{3})R\ 30°$ LEED pattern near $\theta = \frac{1}{3}$ would have lent considerable support to the above interpretation. However, no ordered hydrogen adatom superstructures have been observed on either Ir(111) or Pt(111)[19] near a fractional coverage of one-third. This apparent discrepancy between the LEED observations and the measured energetics of desorption could be reconciled if both hcp and fcc threefold hollow sites are occupied at low (i.e., $\theta \leqslant \frac{1}{3}$) hydrogen coverages.

3. Adsorption kinetics

The adsorption kinetics of hydrogen and deuterium on Ir(111) at 100 K have been characterized by integration of thermal desorption spectra such as those shown in Fig. 2. The integrated area of the mass spectrometric intensity with respect to time is proportional to the adsorbate coverage, and these data (the fractional coverage vs the exposure) represent an integrated form of adsorption kinetics. The coverage dependence (which implies a mechanism) of the adsorption reaction can be deduced by comparing these integrated data with integrated functional forms of various assumed rate expressions, e.g., Langmuir adsorption kinetics[47] and precursor-mediated adsorption kinetics.[48] A quantitative assessment of the adsorption kinetics is dependent upon the assignment of an absolute coverage, which specifies the value of the initial probability of adsorption. We shall assume that the saturation density of hydrogen is equal to the metallic substrate density, i.e., $\theta_H^{sat} = 1$, or $n_H^{sat} = 1.57\times10^{15}$ atoms cm^{-2}. This value is reasonable when compared with values obtained under similar conditions on other hexagonally close-packed surfaces of the Group VIII metals, e.g., 0.8–0.9 on Pt(111)[19,22] and 0.85 ± 0.15 on Ru(001).[42] Since the measurement of the absolute initial probability of adsorption via TDMS is typically accurate to only $\sim 20\%$, this additional uncertainty is of minor importance.

The relative fractional coverage is displayed as a function of exposure in Fig. 5. for (a) hydrogen and (b) deuterium. The solid lines represent the results of least-squares fits of the data to second-order Langmuir adsorption kinetics, i.e., the probability of adsorption is proportional to the square of the fraction of vacant surface sites, $(1 - \theta/\theta^{sat})^2$. The adsorption of both hydrogen and deuterium is well represented by a second-order Langmuir model, and both exhibit, within experimental error, an identical initial probability of adsorption of 7×10^{-3}. This value of the initial probability of adsorption is significantly lower than values obtained on other hexagonally close-packed surfaces of the Group VIII metals, e.g., 0.05 on Ni(111),[15] 0.65 on Rh(111),[40] 0.1–0.25 on Ru(001),[41,42] and 0.016–0.1 on Pt(111).[2,19,20] However it is *identical* to that found for the lower binding energy β_1 adstate on the Ir(110)-(1×2) surface that has been attributed to adsorption on the (111) microfacets of this reconstructed surface.[18] Second-order Langmuir adsorption kinetics have been reported for hydrogen on Rh(111),[40] Ru(001),[41,42] Pt(111)[2,20] and for the β_1 adstate on Ir (110)-(1×2).[18] For small values of the initial probability of adsorption ($s_0 \ll 1$), Kisliuk's *second-order* precursor model[48] reduces to $s(\theta)/s_0 = (1-\theta)^2/$

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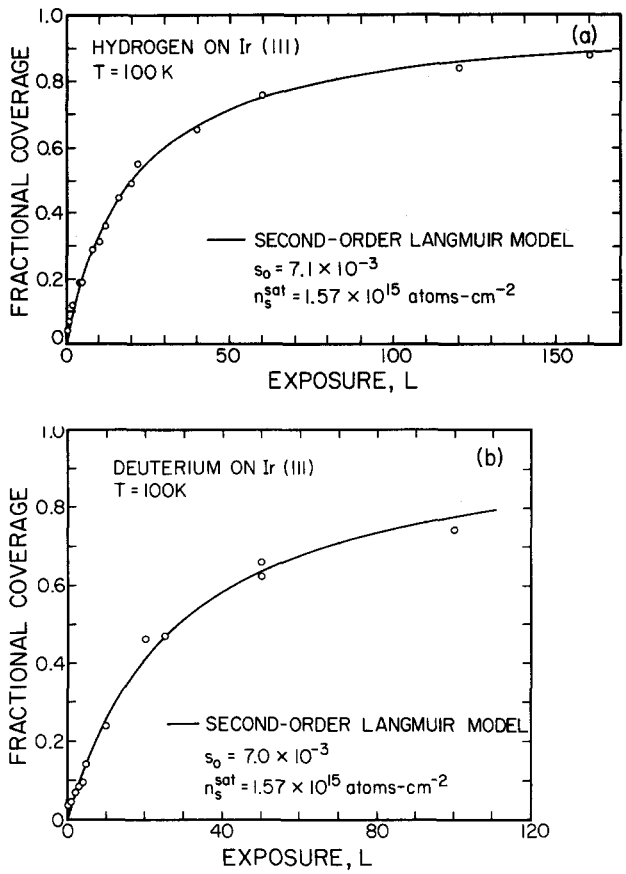


FIG. 5. The coverage–exposure relationship for (a) hydrogen and (b) deuterium on the Ir(111) surface. The solid curves represent least-squares fits of the data to second-order Langmuir kinetics with an initial probability of adsorption of (a) 7.1×10^{-3} for H_2, and (b) 7.0×10^{-3} for D_2.

$[1 + (K-1)\theta]$, where $-s_0 \leqslant K \leqslant \infty$. Thus, if the parameter K is approximately equal to unity, this second-order precursor model becomes equivalent to and experimentally indistinguishable from Langmuir adsorption kinetics. The parameter K is equal to unity if the rates of desorption and migration from the precursor state are independent of the occupation of the underlying site, i.e., except for adsorption, the intrinsic and extrinsic precursor states are equivalent. Since s_0 is much less than unity in this case, we cannot rule out the possibility that the adsorption of hydrogen on Ir(111) proceeds through a weakly bound precursor state for which $K \sim 1$.

Consistent with the desorption kinetics discussed above, the adsorption kinetics also suggest the presence of a small fraction of defects on the Ir(111) surface studied here. These defect sites would be expected to increase the apparent probability of adsorption at low coverages. For example, Christmann and Ertl[3] have observed the initial probability of adsorption of hydrogen to increase by over a factor of 3 when comparing the stepped Pt[9(111)×(111)] surface ($s_0 = 0.34$) to a "perfect" Pt(111) surface ($s_0 \leqslant 0.1$), whereas above a fractional coverage of approximately 0.25, the two surfaces exhibited virtually identical (differential, i.e., $d\theta/d\epsilon$) adsorption kinetics. Examination of the complete data set for the adsorption of hydrogen on this Ir(111) surface reveals that the coverages observed at low exposures ($\epsilon \leqslant 5$ L) are greater than those predicted from the "one-site" adsorption model. This is seen clearly for the low exposure data that are shown in Fig. 6. These data have been fit to a "two-site" adsorption model which is defined by

$$\theta = x_s\theta_s + (1-x_s)\theta_f$$

and

$$\frac{d\theta}{d\epsilon} = x_s s_{0,s}(1-\theta_s)^n + (1-x_s)s_{0,f}(1-\theta_f)^2,$$

where $\epsilon \equiv 2F_{H_2}t/n_s$ is the exposure in units of atoms site^{-1}, x_s is the fraction of defect (e.g., step) sites, and the subscripts "s" and "f" refer to "step" and "flat" (111) sites. If we assume that the defect sites exhibit behavior similar to that observed for the β_2 adstate on the Ir(110)-(1×2) surface,[18] namely an initial probability of adsorption of unity (i.e., $s_{0,s} = 1$), the data are in excellent *quantitative* agreement with the two-site adsorption model by assuming either first- or second-order Langmuir adsorption kinetics for the step sites (i.e., $n = 1$ or 2).[49] This agreement is shown explicitly in Fig. 6, where an optimum fit to the data was obtained with a fraction of defect sites of 0.025. The data for deuterium were not fit to the two-site model since the scatter in these data was greater due to a fewer number of data points for deuterium [e.g., the data for hydrogen displayed in Figs. 5(a) and 6 represent the average values of a number of separate experiments]. In view of the experimental uncertainties in evaluating the defect site density via thermal desorption measurements, the agreement between the values deduced from the adsorption and the desorption kinetics is very reasonable.

We note in passing that knowledge of the relative population of defect sites is often crucial to a more complete understanding of results obtained concerning other fundamental gas–surface interactions, e.g., heterogeneous catalysis.

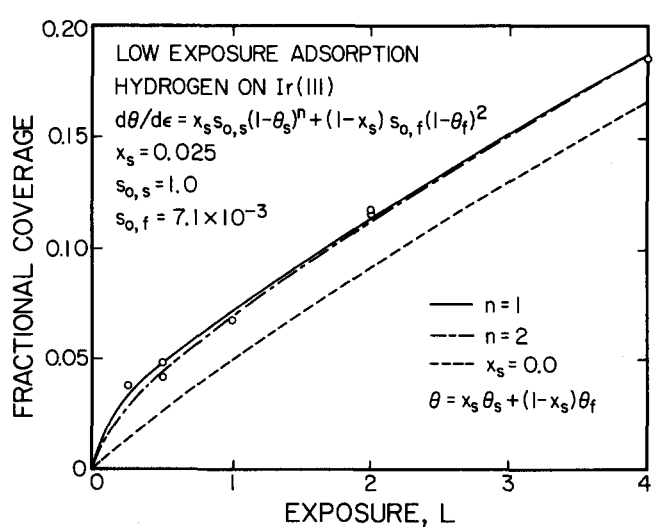


FIG. 6. The coverage–exposure relationship for low exposures of hydrogen to the Ir(111) surface. In this coverage regime, the "two-site" adsorption model (i.e., the fraction of defect sites $x_s = 0.025$, and $n = 1$ or 2; see the text) provides a superior fit of the data with respect to the "one-site" model (i.e., $x_s = 0.0$) (Ref. 49).

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For example, the product distribution from the hydrogenolysis of *n*-butane over iridium surfaces has been found to depend sensitively on the concentration of step sites (low-coordination-number metal surface atoms),[50] whereas these same sites are apparently also necessary for the relatively facile activation of C–H bonds in saturated hydrocarbons on iridium.[51]

B. Hydrogen on Pt(110)-(1×2)

1. LEED observations

As in the case of the Ir(111) surface, no ordered adsorbate superstructures were observed under any conditions examined, and the (1×2) LEED pattern representative of the clean Pt(110) surface was always present. These observations are in agreement with those reported previously for the adsorption of hydrogen on the structurally similar Ir(110)-(1×2) surface.[18]

2. Desorption kinetics

Thermal desorption spectra of D_2 from Pt(110)-(1×2) for nine different initial coverages are shown in Fig. 7. The linear heating rate produced by employing the digital temperature controller for each spectrum was 3.0 K s^{-1}. As for the Ir(111) surface, hydrogen and deuterium produce thermal desorption spectra that are identical within experimental error. Unlike the Ir(111) surface, however, two distinct desorption peaks, or "states", are clearly apparent in Fig. 7. Previous work[2,52] did not resolve these two desorption features clearly due to the high heating rates that were employed in those studies ($\beta \geqslant 40$ K s^{-1}). The state desorbing at approximately 300 K, which we label the β_2 adstate (consistent with previous work on the Ir(110)-(1×2) surface,[18] saturates at low exposures (i.e., a high probability of adsorption). The narrow width of the thermal desorption peaks from this β_2 adstate (FWHM ~ 30 K; compare Fig. 7 with Fig. 2) suggests the presence of *attractive* adatom–adatom interactions.[39] The state desorbing near 200 K, which we label as the β_1 adstate, saturates at much larger exposures compared to the β_2 adstate, and the peak temperature decreases continuously with increasing coverage, indicative of second-order desorption. Broadening of the desorption peaks of the β_1 adstate is apparent also, quite similar to that observed on Ir(111) (cf. Fig. 2), and this suggests the presence of repulsive adatom–adatom interactions. The ratio of the saturation densities of the two adstates, β_2:β_1, is approximately 1:2. These thermal desorption spectra are *very* similar to those obtained on the Ir(110)-(1×2) surface.[18]

Deuterium and hydrogen were adsorbed sequentially to determine the approximate extent of exchange between the two adstates. Thermal desorption spectra of D_2, HD, and H_2 (not corrected for differences in pumping speeds or mass spectrometric sensitivities), obtained at a heating rate of 3.0 K s^{-1} following sequential exposures of 0.64 L of D_2 and 8.5 L of H_2, are shown in Fig. 8. This exposure of D_2 populates approximately 85% of the β_2 adstate, whereas the additional exposure of H_2 fills the remainder of the β_2 adstate and populates approximately 35% of the β_1 adstate (see Fig. 11 below). Thus, the initial coverage of hydrogen adatoms and deuterium adatoms is approximately equal. There is significant exchange between the two adstates during the course of the TDMS measurements, occurring at temperatures as low as 200 K. Similar thermal desorption spectra are obtained if the exposure sequence is reversed. These results are in

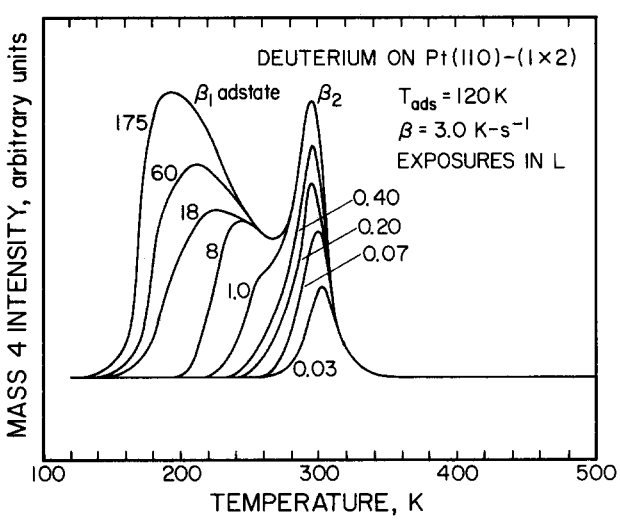


FIG. 7. Thermal desorption spectra of deuterium from the Pt(110)-(1×2) surface parametric in exposure of deuterium. The surface temperature during adsorption was 120 K, and the linear heating rate for each spectrum was 3.0 K s^{-1}. The ratio of the saturation densities of the two adstates, β_2:β_1, is approximately 1:2.

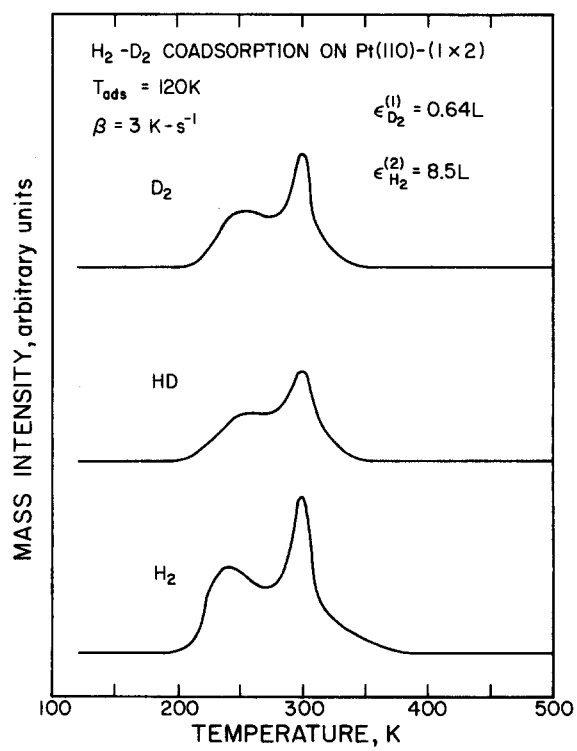


FIG. 8. Thermal desorption specta of D_2, HD, and H_2 from a co-adsorbed overlayer of hydrogen and deuterium. In this experiment, 0.64 L of D_2 was exposed first, populating approximately 85% of the β_2 adstate, and, subsequently, 8.5 L of H_2 was exposed. The surface temperature during adsorption was 120 K, and the linear heating rate was 3.0 K s^{-1}.

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(qualitative) agreement with results obtained from similar isotopic exchange experiments carried out on the Ir(110)-(1×2) surface.[18]

The desorption kinetics were characterized quantitatively by employing the integral method involving variable heating rates described above. In this case, the heating rates were varied from 2 to 25 K s^{-1}. Three to four spectra were measured at each heating rate and summed to increase the signal-to-noise ratio. Two different initial coverages, $\theta_0 = 0.37$ and $\theta_0 = 0.70$, were employed in order to examine two different coverage regimes: in this case, the β_2 and β_1 adstates. In particular, the lower initial coverage was utilized to examine fractional coverages below approximately 0.35, and the higher initial coverage was employed to examine fractional coverages between approximately 0.35 and 0.65. We note that *serious* errors could arise if the *entire* coverage range were examined by utilizing a single high initial coverage, since two distinct states are clearly apparent in this case. These errors would arise from the decreasing resolution of the two desorption features at high heating rates ($\beta \gg 3$ K s^{-1}), as observed in previous work.[2,52] Since desorption from the β_1 adstate is characterized by a lower activation energy (lower desorption temperature), the β_1 peak will shift upward in temperature more than the β_2 peak as the heating rate is increased, resulting in a mixing of the two desorption features at intermediate coverages (i.e., $0.2 \lesssim \theta \lesssim 0.45$). This can be avoided, at least for desorption from the β_2 adstate, by employing an initial coverage representative of (essentially) only the β_2 adstate, i.e., $\theta_0 = 0.37$.

Making use of data from Arrhenius plots similar to those shown in Fig. 3, the activation energy of desorption and the preexponential factor in the desorption rate coefficient have been evaluated as a function of surface coverage. These results are shown in Fig. 9. The preexponential factor was evaluated by assuming that the saturation density of hydrogen is 1.60×10^{15} atoms cm^{-2} (see Sec. III B 3) and that the desorption reaction is second order. The rate parameters are strongly dependent on coverage for desorption

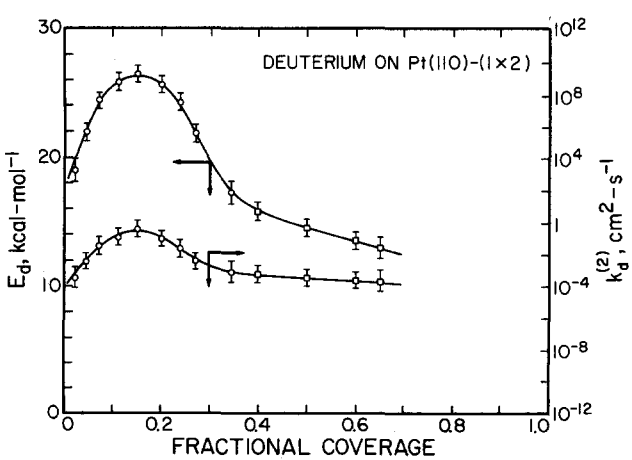


FIG. 9. Activation energy (E_d) and second-order preexponential factor ($k_d^{(2)}$) of the desorption rate coefficeint for D_2 desorption from Pt(110)-(1×2) as a function of fractional surface coverage.

from the β_2 adstate. In addition, a compensation effect is clearly apparent for desorption from the β_2 adstate, both rate parameters first increasing and then decreasing with coverage. In particular, from a zero-coverage limit of approximately 18 kcal mol^{-1}, the activation energy of desorption first increases to a value of 26.5 kcal mol^{-1} at a fractional coverage of 0.15 and then decreases to a value of approximately 18 kcal mol^{-1} as the coverage increases to a value of 0.32, i.e., saturation of the β_2 adstate. Concomitantly, from a zero-coverage limit of approximately 10^{-4} cm^2 s^{-1}, the preexponential factor of the desorption rate coefficient first increases to a value of 0.3 cm^2 s^{-1} and then decreases to a value of approximately 10^{-3} cm^2 s^{-1}. The approximate symmetry of the rate parameters about a fractional coverage of $\theta \sim 0.15$ (i.e., one-half of saturation of the β_2 adstate) should be noted.

The initial increase in the activation energy of desorption as the fractional coverage increases to a value of 0.15 is due to attractive adatom–adatom interactions at these adsorbate concentrations. The presence of attractive interactions in the adlayer was implied by the narrow width of the β_2 adstate desorption peak. These interactions act to increase the (average) binding energy of the hydrogen adatoms and, hence, the activation energy of desorption since there is no activation energy of adsorption for the β_2 adstate (adsorption probability near unity; see Sec. III B 3). An initial increase in the activation energy of desorption has been observed also for hydrogen on the (110) surfaces of both iridium[18] and nickel (somewhat less pronounced).[14] The fact that these attractive interactions are much more apparent on fcc (110) surfaces, compared to fcc (111) surfaces, for example, implies that surface structural effects are important.

The (110) surfaces of fcc metals consist of close-packed rows of atoms separated by one-dimensional channels (see Fig. 1). The probable location of the low coverage β_2 adstate of hydrogen is the pseudo-fourfold hollow "trough" sites on the Ir(110)-(1×2) surface.[18] The saturation density of the β_2 adstate on the Pt(110)-(1×2) surface is approximately equal to the density of these trough sites, i.e., one hydrogen adatom per fourfold hollow site at saturation. It appears that these one-dimensional channels can mediate particularly well "through lattice" attractive adatom–adatom interactions between *next*-nearest-neighbors along the [001] channel direction. From this point of view, the decrease in the activation energy of desorption between fractional coverages of approximately 0.15 and 0.30 is due to the occurrence of *repulsive* adatom–adatom interactions between *nearest* neighbors. If both attractive next-nearest-neighbor and repulsive nearest neighbor interactions occur, one might expect to observe ordering in one dimension along the [001] direction at a fractional coverage corresponding to one-half saturation of the β_2 adstate (i.e., $\theta = 0.15$). However, if the interaction *between* adjacent channels were sufficiently weak that ordering in *two* dimensions did not occur (i.e., no correlation between the one-dimensional channels), one would not expect to observe a new LEED pattern different from the clean surface (1×2) pattern. Since no new LEED patterns were observed, we can exclude *two*-dimensional or-

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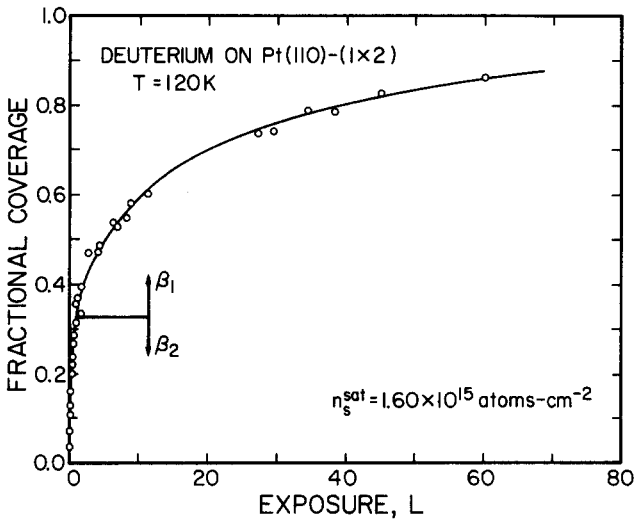


FIG. 10. The coverage-exposure relationship for deuterium on the Pt(110)-(1×2) surface. The solid curve has been drawn to facilitate the presentation. The coverage regimes representative of the β_2 and β_1 adstates are marked to emphasize the vastly different kinetics exhibited by the two states.

dering in the β_2 adstate, whereas *one*-dimensional ordering remains a distinct possibility.

The coverage dependence of the kinetic rate parameters of the β_1 adstate is also apparent from an inspection of Fig. 9. The activation energy of desorption decreases approximately linearly with increasing coverage from a value of 17 kcal mol^{-1} at $\theta = 0.35$ to a value of 13 kcal mol^{-1} at $\theta = 0.65$, whereas the preexponential factor maintains an essentially constant value of approximately 3×10^{-4} cm^2 s^{-1}. The decrease of the activation energy is again due to the occurrence of repulsive adatom-adatom interactions which act to decrease the binding energy of the hydrogen adatoms. It is important to note that desorption from the β_1 adstate occurs in the presence of a fully occupied β_2 adstate. Thus, in addition to interactions *within* the β_1 adstate, interactions *between* the β_1 and β_2 adstates may contribute also to the repulsive energy which leads to a decrease in the activation energy of desorption from the β_1 adstate.

3. Adsorption kinetics

The adsorption kinetics of hydrogen and deuterium on Pt(110)-(1×2) at 120 K have been characterized by integration of thermal desorption spectra similar to those shown in Fig. 7. The fractional coverage of deuterium adatoms is displayed as a function of exposure in Fig. 10, and hydrogen exhibits identical adsorption kinetics. The absolute coverage has been assigned by comparing the integrated intensities of thermal desorption spectra corresponding to saturated overlayers at 120 and at 170 K. The saturation density of deuterium adatoms at 170 K has been measured to be $1.07 \pm 0.1\times10^{15}$ atoms cm^{-2}.[36] We find that the saturation density at 120 K is 1.60×10^{15} atoms cm^{-2}. As may be seen in Fig. 10, the adsorption kinetics are represented by two distinct regimes corresponding to the β_2 and β_1 adstates. Since the two adstates exhibit vastly different adsorption kinetics, they can be examined separately.

The fractional coverages, relative to each of the adstate saturation values, are displayed as a function of exposure in Fig. 11 for (a) the β_2 and (b) the β_1 adstate ($n_H^{sat} = 0.52\times10^{15}$ atoms cm^{-2} for the β_2 adstate, and $n_H^{sat} = 1.08\times10^{15}$ atoms cm^{-2} for the β_1 adstate). The solid lines represent the results of least-squares fits of the data to (a) first-order and (b) second-order Langmuir adsorption kinetics. The adsorption of deuterium into the β_2 adstate is characterized by an initial probability of adsorption of 0.45, whereas the corresponding probability for the β_1 adstate is considerably smaller, 0.024. Analysis of the adsorption kinetics of hydrogen results in values for the initial probability of adsorption of 0.46 for the β_2 adstate and 0.022 for the β_1 adstate, i.e., experimentally indistinguishable from deuterium.

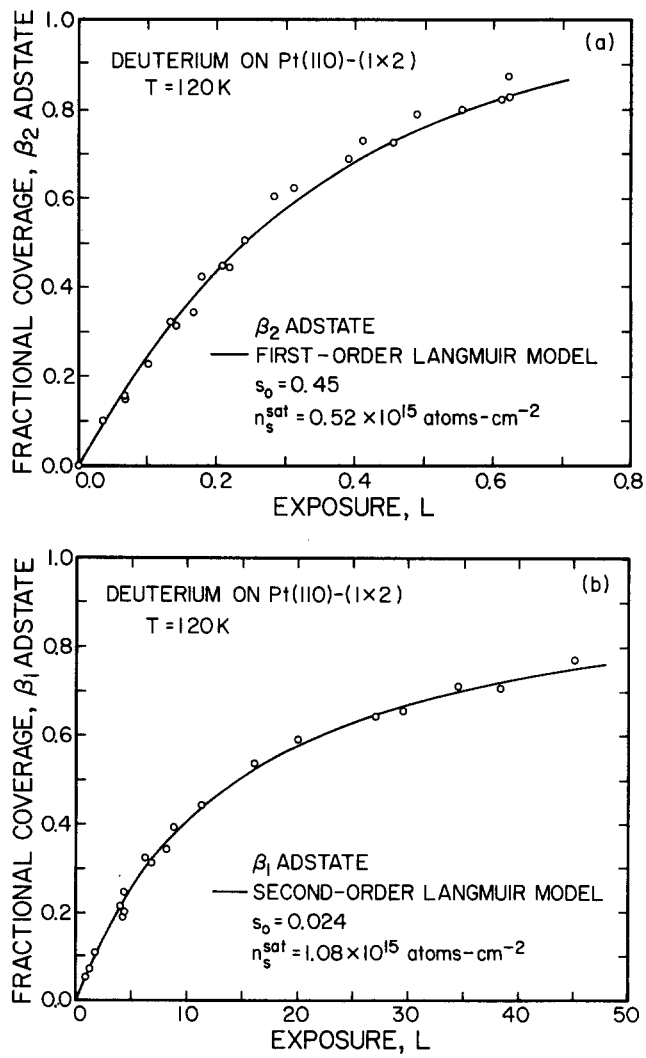


FIG. 11. Analytic fits to the adsorption kinetics of D_2 on the Pt(110)-(1×2) surface derived from the data shown in Fig. 10. (a) The adsorption kinetics for D_2 into the β_2 adstate to fit to a first-order Langmuir model with an initial probability of adsorption of 0.45. The coverage has been normalized to saturation of the β_2 adstate. (b) The adsorption kinetics for D_2 into the β_1 adstate fit to a second-order Langmuir model with an initial probability of adsorption of 0.024. The coverage has been normalized to saturation of the β_1 adstate.

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Since the adsorption of hydrogen is dissociative, the observation of *first*-order Langmuir adsorption kinetics $[s(\theta/\theta^{sat})/s_0 = (1-\theta/\theta^{sat})]$ for the β_2 adstate is somewhat unexpected. However, the β_2 adstate on the structurally similar Ir(110)-(1×2) surface exhibits first-order Langmuir adsorption kinetics.[18] First-order adsorption kinetics have been reported also for hydrogen on the (111) and (100) surfaces of nickel.[14] The initial probability of adsorption of 0.46 for the β_2 adstate compares well to both the value of 0.33 reported previously by Lu and Rye[2] for the Pt(110) surface and the value of 0.34 attributed to adsorption at step sites on the Pt[9(111)×(111)] surface.[3]

As for the β_1 adstate on Pt(110)-(1×2), second-order Langmuir adsorption kinetics have been reported both for the β_1 adstate on the Ir(110)-(1×2) surface, as well as for the hexagonally close-packed surfaces of Rh,[40] Ru,[41,42] Pt,[2,20] and Ir (this work). The value of 0.022 for the initial probability of adsorption compares well to the value of 0.016 reported by Lu and Rye for hydrogen adsorption on Pt(111),[2] supporting the proposition that, as for the Ir(110) − (1×2) surface,[18] the β_1 adstate can be attributed to adsorption on the (111) microfacets of the reconstructed Pt(110) − (1×2) surface. Finally, as for the Ir(111) surface, second-order Langmuir adsorption kinetics, characterized by a small initial probability of adsorption, cannot be distinguished by these experiments from precursor-mediated kinetics in which the precursor state interacts weakly with the surface.

4. Contact potential difference measurements

Contact potential difference measurements have been carried out in order to evaluate the work function change ($\Delta\varphi$) upon adsorption of hydrogen on the Pt(110)-(1×2) surface. These measurements can provide information concerning electron transfer between the adsorbate and the substrate, i.e., the dipole moment of the adsorbate–substrate complex. For the data presented here, the CPD has been measured as a function of time while the surface was exposed to a constant partial pressure of hydrogen. Since the adsorption kinetics of hydrogen on the Pt(110)-(1×2) surface are characterized by two distinct regimes, a relatively lower partial pressure was employed to examine the CPD for the β_2 adstate (approximately 2×10^{-8} Torr), whereas a higher one was employed for the β_1 adstate (approximately 5×10^{-7} Torr). Making use of the coverage vs exposure relationships shown in Figs. 11(a) and 11(b), the CPD was evaluated as a function of coverage.

The CPD for hydrogen adsorbed on the Pt(110)-(1×2) surface at 120 K is shown in Fig. 12 as a function of the (total) fractional coverage. Adsorption into the β_2 adstate results in an increase in the CPD, reaching a maximum of 0.17 eV at a coverage of 0.3. As the coverage increased beyond 0.3, characteristic of adsorption into the β_1 adstate, the CPD decreases, approaching a value of − 0.50 eV at saturation. For coverages below approximately 0.2 and above approximately 0.4, the CPD varies linearly with coverage, indicating that the dipole moment of the adsorbed hydrogen adatoms is constant at these concentrations (i.e., depolarization effects are small). The calculated values for the dipole moments (μ) are + 0.12 D for the β_2 adstate ($\theta \leqslant 0.2$), and − 0.17 D for the β_1 adstate ($\theta \geqslant 0.4$), where 1 D ≡ 1 esu cm. The nonlinear variation in the CPD at the intermediate coverages between 0.2 and 0.4 is due primarily to the partial filling of the β_1 adstate prior to the filling of the β_2 adstate, and, possibly, the occurrence of depolarization effects within the β_2 adstate. Note that depolarization effects are absent in the β_1 adstate as the coverage approaches saturation.

An initial *increase* in the CPD, followed by a continuous *decrease* has been observed for hydrogen absorbed on both the Ir(110)-(1×2) surface[18] and the stepped Pt[9(111)×(111)] surface.[3] On the structurally similar Ir(110)-(1×2) surface, the CPD increases linearly to a maximum of 0.30 eV at a coverage of 0.33 (μ = + 0.14 D), corresponding to the filling of the higher binding energy β_2 adstate. As the lower binding energy β_1 adstate populates, the CPD decreases and approaches a saturation value of − 0.14 eV. On Pt[9(111)×(111)], the CPD increases to a maximum of 0.02 eV at a coverage of 0.25, and then decreases continuously to a value of − 0.36 eV at saturation coverage. For fractional coverages above 0.7, the CPD of this stepped platinum surface becomes indistinguishable from that obtained on the Pt(111) surface, where the CPD is observed to decrease continuously from zero coverage.[19,21]

The apparent disparity in the sign of the dipole moment for the low coverage, higher binding energy states {the β_2 adstates on the (110)-(1×2) surfaces of Pt and Ir, and the step sites on Pt[9(111)×(111)]} and the lower binding energy states [the β_1 adstates and/or the (111) "terrace sites"] can be explained by the following arguments. Clearly, the continuous decrease in the CPD observed on Pt(111)[19,21] is due to electron donation from the hydrogen adatoms to the metal surface. Since we attribute the β_1 adstates to adsorption on the (111) microfacets of the (110)-(1×2) surfaces, this explanation holds also for the observed decrease in the CPD for the filling of the β_1 adstates. The increase in the CPD for the β_2 adstates can be interpreted as

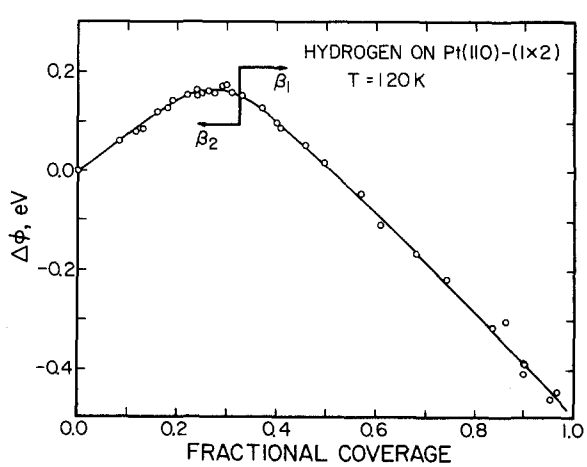


FIG. 12. The CPD of hydrogen on the Pt(110)-(1×2) surface as a function of fractional coverage at 120 K. The coverage regimes representative of the β_2 and β_1 adstates are marked to emphasize the (apparently) different charge transfer characteristics of the two adstates.

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either (i) electron donation from the metal surface to the hydrogen adatoms, assuming the adatoms lie above the image dipole plane of the surface, or (ii) electron donation from the hydrogen adatoms to the metal surface (as for the β_1 adstates) with the adatoms lying *below* the image plane (e.g., the pseudo-fourfold hollow trough sites, cf., Fig. 1). The latter viewpoint, the β_2 adstate being located in the troughs, was adopted previously to explain the CPD behavior on the Ir(110)-(1×2) surface on the basis of adsorption-induced changes in the UV photoelectron spectra.[18] We believe this interpretation is also applicable to the results obtained here on the Pt(110)-(1×2) surface {and elsewhere on the Pt[9(111)×(111)] surface,[3] adsorption at the inner corner of the steps being below the image plane[54]}. Indeed, the proposition of the pseudo-fourfold hollow adsite for the β_2 adstate is consistent with the observation that hydrogen tends to absorb preferentially at high coordination surface sites.[1]

5. Co-adsorption of CO and H_2

The co-adsorption of hydrogen and carbon monoxide on the Pt(110)-(1×2) surface was studied in order to examine the perturbations by CO on the adsorption and desorption of hydrogen. It is of interest to review briefly the results obtained previously concerning the co-adsorption of CO and H_2 on the structurally similar Ir(110)-(1×2) surface.[55] The adsorption of hydrogen on an Ir(110)-(1×2) surface exposed previously to CO results in less hydrogen occupying the high binding energy β_2 adstate, and occupation of the β_2 adstate at a lower surface coverage of hydrogen. The postadsorption of CO on a hydrogen precovered surface causes a preferential shift of the hydrogen from the β_2 to the β_1 adstate. In both cases, the adsorption of one-half of a monolayer of CO poisons or displaces completely all of the hydrogen chemisorbed in the β_2 adstate, shifting the occupancy to the β_1 adstate.

Thermal desorption spectra of H_2 from Pt(110)-(1×2), corresponding to the same initial coverage of hydrogen ($\epsilon_{H_2} = 0.39$ L, $\theta_H = 0.26$, i.e., 80% of saturation of the β_2 adstate) and six different subsequent exposures of CO at 120 K, are shown in Fig. 13(A) ($\theta_{CO} = 1$ is saturation and is equal to 0.92×10^{15} molecule cm^{-2} [53]). Since the linear heating rate was 15 K s^{-1}, the spectra are shifted to slightly higher temperatures with repect to those shown in Fig. 7 for the clean surface (where $\beta = 3.0$ K s^{-1}). In all cases, H_2 desorption was complete prior to any desorption of CO ($T \geqslant 400$ K). For exposures of CO below approximately 1 L, desorption from the β_2 adstate is suppressed progressively with increasing CO exposures, whereas desorption between 200 and 300 K (characteristic of the β_1 adstate on the clean surface) increases concomitantly. For exposures above 1 L, the H_2 thermal desorption peak downshifts further in temperature to approximately 200 K. Integration of these thermal desorption spectra indicates that displacement of hydrogen from the surface at 120 K has not occurred for these exposures of CO. In addition, no new LEED patterns were observed for these exposure sequences for temperatures between 120 and 300 K.[56] This observation, coupled with the fact that H_2 desorption from the β_2 adstate is extinguished completely at fractional coverages of CO below 0.5, indicates that (long rang) adsorbate segregation is not occurring, which, for example, does occur on the Rh(111) surface.[58]

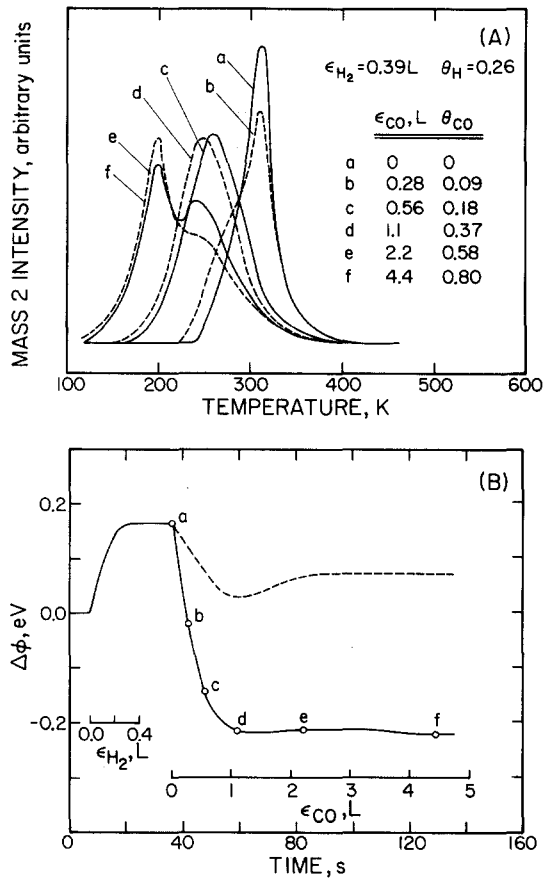


FIG. 13. (A) Thermal desorption of H_2 from preadsorbed hydrogen (0.39 L; $\theta_H = 0.26$) exposed to varying amounts of CO (0–4.4 L CO; $\theta_{CO} = 0$–0.8). The surface temperature during adsorption was 120 K, and the linear heating rate was 15 K s^{-1}. The coverage of hydrogen was constant with these CO exposures. (B) The CPD for the exposure sequence represented by the thermal desorption spectra given in (A). The surface was exposed to 0.39 L of H_2, followed by a constant partial pressure of CO of 5×10^{-8} Torr. The dashed line in (B) represents the CPD for CO exposed to a *clean* Pt(110)-(1×2) surface.

Three desorption regimes, dependent upon the CO postexposure, are apparent in Fig. 13(A). At zero postexposure, desorption is exclusively from the β_2 adstate. For postexposures between approximately 0.5 and 1 L, the β_2 adstate has been virtually extinguished, and the desorption peaks are very similar to those expected if the hydrogen adatoms were displaced [i.e., no additional CO(a)–H(a) interactions] to the β_1 adstate (c.f., Fig. 7 and $\epsilon_{D_2} = 8$ L). For postexposures above 1 L, this "β_1-like" adstate is suppressed, and a new "state" with a peak temperature of approximately 200 K begins to dominate the desorption spectra. If exposures below ~1 L displace completely hydrogen adatoms to the β_1 adstate, it is reasonable to assume that exposures above 1 L ($\theta_{CO} \geqslant 0.4$) may lead to repulsive interactions between the CO admolecules and the hydrogen adatoms. These interactions would explain the observed splitting and shift in the desorption peak to lower temperature.

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Contact potential difference measurements were carried out to test the above hypothesis concerning the effects of coadsorbed CO on the chemisorption of hydrogen. Since the two clean surface adstates of hydrogen are characterized by dipole moments of opposite sign (c.f., Fig. 12), displacement of adatoms from one adstate to the other should result in large changes in the CPD, whereas (nondisplacing) interactions between the CO admolecules and the hydrogen adatoms may only marginally affect the CPD (assuming we can account for the CPD that is due solely to the CO). An experiment was conducted in order to measure the CPD for the exposure sequence represented by the thermal desorption spectra shown in Fig. 13(A). Specifically, the CPD was measured as a function of time while the clean Pt(110)-(1×2) surface was exposed to 0.39 L of hydrogen, followed by a constant partial pressure of CO of 5×10^{-8} Torr. These data are shown in Fig. 13 (B). The CPD for CO adsorbed on the clean surface is shown also for reference purposes. (Within experimental error, the adsorption kinetics of CO were found to be identical for the clean surface and for this precoverage of hydrogen.) From Fig. 13 (B) we see that the CPD decreases precipitously upon the initial exposure of CO, decreasing to ~ -0.02 eV at an exposure of 0.3 L and to ~ -0.21 eV at 1L. Above exposures of approximately 1 L, the CPD is essentially constant at ~ -0.22 eV.

These results for the CPD are entirely consistent with the above interpretation concerning the thermal desorption spectra. For exposures below approximately 1 L, the CPD decreases rapidly as the hydrogen adatoms are displaced to the β_1 adstate. However, for exposures above approximately 1 L, little change in the CPD is observed, consistent with the occurrence of (nondisplacing) interactions between the CO admolecules and the hydrogen adatoms which lead to the observed changes in the thermal desorption spectra.

The effect of preadsorbed CO on the chemisorption of hydrogen on the Pt(110)-(1×2) surface was studied briefly. It was found that an exposure of 1.1 L of CO ($\theta_{CO}\sim0.4$) blocked completely the adsorption of H_2 into the β_2 adstate. The resulting adsorption into the β_1 adstate was characterized by an initial probability of adsorption of 0.016 and second-order Langmuir kinetics, i.e., very similar to the clean surface on which $s_0 = 0.022$ and second-order Langmuir kinetics applied also. The thermal desorption spectra from this CO precovered surface were found to be similar to those expected for hydrogen adatoms which were merely displaced to the β_1 adstate, e.g., spectrum d in Fig. 13(A). Finally, adsorption of hydrogen on this CO precovered surface resulted in a continuous decrease in the CPD, consistent with preferential adsorption into the β_1 adstate.

In summary, the post- or preadsorption of CO on the Pt(110)-(1×2) surface poisons hydrogen adsorption into the β_2 adstate and shifts its occupation to the β_1 adstate, consistent with earlier work on the structurally similar Ir(110)-(1×2) surface.[55] At low to moderate coverages of CO ($\theta_{CO}\leqslant0.4$) the CO acts merely to displace the hydrogen adatoms from the β_2 to the β_1 adstate. However, at CO coverages above one-half of a monolayer, repulsive interactions between the CO admolecules and the hydrogen adatoms lead to a reduced binding energy of the hydrogen adatoms and, hence, a lower desorption temperature of hydrogen.

IV. DISCUSSION

The results obtained here concerning the chemisorption of hydrogen on both the Ir(111) and Pt(110)-(1×2) surfaces illustrate clearly the importance of surface structure and its influence on the rates of both adsorption and desorption. The initial probability of adsorption can vary by as much as two orders of magnitude when comparing the close-packed (111) and the corrugated (110)-(1×2) surfaces of both iridium and platinum. However, the adsorption kinetics characteristic of the lower binding energy β_1 adstates on the (110)-(1×2) surfaces are virtually identical to those observed on the corresponding (111) surfaces. The desorption kinetics are also influenced strongly by the surface structure. The (110)-(1×2) surfaces possess unique adsites as a consequence of their geometry, and these sites are related directly to the observation of an additional [with respect to the (111) surfaces] higher binding energy β_2 adstate. Furthermore, these (110)-(1×2) surfaces apparently mediate particularly well attractive adatom–adatom interactions between the hydrogen adatoms in this higher binding energy β_2 adstate. As in the case of the adsorption kinetics, the desorption kinetics characteristic of the lower binding energy β_1 adstates on the (110)-(1×2) surfaces are very similar to those observed on the corresponding (111) surfaces. In the discussion that follows, we shall quantify these comparisons of the adsorption and desorption kinetics of

TABLE I. Hydrogen adsorption kinetics on iridium and platinum.[a]

	Iridium			Platinum		
	s_0	$f(\theta)$	Ref.	s_0	$f(\theta)$	Ref.
(111)	7×10^{-3}	$(1-\theta)^2$	b	0.016	$(1-\theta)^2$	2, 59
(110)-(1×2) β_1 adstate	7×10^{-3}	$(1-\theta)^2$	18	0.022	$(1-\theta)^2$	b
(110)-(1×2) β_2 adstate	1.0	$(1-\theta)$	18	0.46	$(1-\theta)$	b

[a] In all cases, the kinetics were evaluated at surface temperatures of 100 to 150 K, and for gas temperatures of approximately 300 K.
[b] This work.

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hydrogen on the (111) and (110)-(1×2) surfaces of both iridium and platinum.

The adsorption kinetics of hydrogen on the (111) and (110)-(1×2) surfaces of both iridium and platinum are summarized in Table I in terms of the initial probabilities of adsorption (s_0) and the functional dependence of the reaction on the fractional coverage [$f(\theta)$].[59] In all cases, the kinetics were evaluated at surface temperatures between 100 and 150 K, and for gas temperatures of approximately 300 K. The dominant role of the surface structure in determining the rate of the adsorption reaction is clearly apparent. Adsorption into the β_2 adstates of both iridium and platinum is characterized by near unity initial probabilities of adsorption and first-order Langmuir adsorption kinetics. These values of s_0 imply that adsorption into the β_2 adstates is not activated.

The importance of the *local* surface structure is supported by the observation of both essentially identical values of the initial probabilities of adsorption and identical second-order Langmuir kinetics for adsorption on the (111) surfaces and into the β_1 adstates on the (110)-(1×2) surfaces. The striking similarity of the adsorption kinetics for these two adstates on macroscopically distinct surfaces is due to the presence of structurally similar threefold hollow adsites. This result was not totally unexpected; however, it is both surprising and provocative that (111) microfacets which are only three atomic rows in width exhibit essentially identical rates of adsorption as a "perfect" (111) surface. The low values observed for the initial probabilities of adsorption on the (111) surfaces could be explained by the presence of an activation barrier for dissociative adsorption. Previous work employing molecular beam techniques has suggested that this barrier is on the order of 0.4–1.5 kcal mol^{-1} on the Pt(111) surface.[7,8] As discussed above, more recent work on the adsorption of hydrogen on the Ni(111) surface[16] implicates the presence of an activation barrier of approximately 2 kcal mol^{-1} on this structurally similar close-packed surface. In addition, in agreement with our point of view [i.e., unactivated adsorption into the low-coverage β_2 adstates on the (110)-(1×2) surfaces], adsorption on the Ni(110) surface was found to be essentially unactivated.

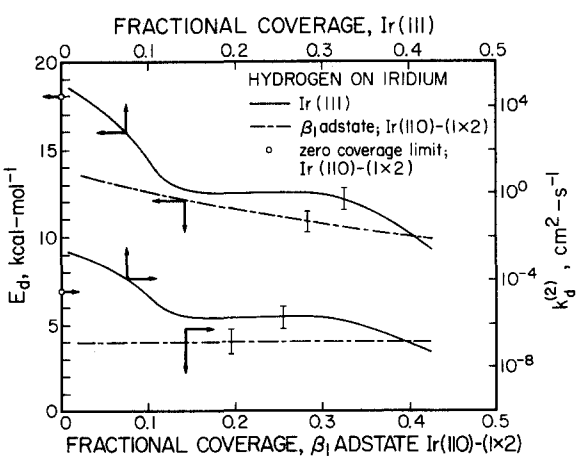


FIG. 14. Comparison of the desorption rate parameters, the activation energy (E_d) and the second-order preexponential factor ($k_d^{(2)}$), for hydrogen on the Ir(111) surface and for the β_1 adstate on the Ir(110)-(1×2) surface as a function of fractional coverage. The abscissae have been scaled such that the absolute adsorbate concentrations are equal for the two adstates. The zero-coverage limit for the Ir(110)-(1×2) surface is shown for reference purposes. The data for the Ir(110)-(1×2) surface are from Ibbotson *et al.* (Ref. 18).

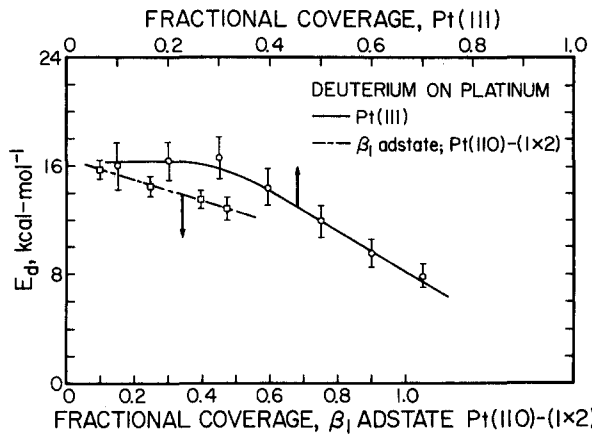


FIG. 15. Comparison of the activation energies of desorption (E_d) for deuterium on the Pt(111) surface and for the β_1 adstate on the Pt(111)-(1×2) surface as a function of fractional coverage. The abscissae have been scaled such that the absolute adsorbate concentrations are equal for the two adstates. The data for the Pt(111) surface are from Norton *et al.* (Refs. 22 and 60).

The kinetic parameters which describe the desorption reaction (the activation energy of desorption and the second-order preexponential factor of the desorption rate coefficient) are displayed as a function of surface coverage in Fig. 14 for iridium and Fig. 15 for platinum (activation energy only)[60] for both the (111) surface and the β_1 adstate on the (110)-(1×2) surface. The abscissae have been scaled such that the *absolute* adorbate concentrations are equal for the two adstates on any vertical line.

For coverages above approximately 0.1, the kinetic parameters for each adstate on the iridium surfaces are given by similar values. However, in the limit of zero coverage, the kinetic parameters characteristic of this Ir(111) surface are closer to the zero-coverage limit on the Ir(110)-(1×2) surface, which is characteristic of the β_2 adstate. This observation supports the proposition argued above that the desorption kinetics at coverages below 0.1 are dominated by the presence of defect sites (e.g., step sites) participating in the desorption reaction. Consequently, the zero-coverage limit on a perfect Ir(111) surface might be expected to be characterized by the (coverage-independent) values measured here for coverages between 0.1 and 0.3, i.e, $E_d \simeq 12.6$ kcal mol^{-1} and $k_d^{(2)} \simeq 2\times10^{-6}$ cm^2 s^{-1}. Note that the former is close to the value of ~ 13.5 kcal mol^{-1} for the zero-coverage limit of the β_1 adstate on the Ir(110)-(1×2) surface.

For fractional coverages above approximately one-third of a monolayer, the activation energy of desorption on the Ir(111) surface begins to decrease. We believe that this is due to the occurrence of repulsive adatom–adatom interactions between nearest neighbors which decrease the (average) binding energy of the hydrogen adatoms and, hence,

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the activation energy of desorption. However, for the β_1 adstate on the Ir(110)-(1×2) surface, a continuous decrease in the activation energy is observed. This result could be explained also by the occurrence of repulsive interactions between the adatoms. For example, since desorption from the β_1 adstate occurs in the presence of a fully occupied β_2 adstate, the two adstates being essentially nearest neighbors (cf., Fig. 1), repulsive interactions between adatoms in the β_1 adstate and those in the β_2 adstate could occur.

As may be seen in Fig. 15, on the platinum surfaces, the activation energies of desorption vary with the surface coverage in a manner similar to that observed on the iridium surfaces. For fractional coverages below 0.3, the activation energy of desorption on the Pt(111) surface is essentially independent of the coverage. In addition, although it was not investigated quantitatively, Nortan *et al.*[22] found evidence for hydrogen that was more strongly bound at coverages below approximately 0.05, presumably due to defect sites. As the coverage increases beyond 0.3, the activation energy of desorption decreases due to the occurrence of repulsive interactions between nearest neighbor hydrogen adatoms. For the β_1 adstate on the Pt(110)-(1×2) surface, the activation energy of desorption decreases continuously with increasing coverage, indicative of the occurrence of repulsive interactions at all adsorbate concentrations for which the β_2 adstate probably plays a role (e.g., via a direct interaction with the β_1 adstate). In the limit of zero coverage, the two adstates exhibit a similar activation energy of desorption of approximately 16 kcal mol^{-1}, emphasizing again the similarity of the two adstates.

The kinetic parameters which describe the desorption of hydrogen from the (110)-(1×2) surfaces of both iridium and platinum are displayed in Fig. 16 as a function of surface coverage. The dominant influence of the surface structure is clearly apparent. For both surfaces, the parameters vary sympathetically with surface coverage, first increasing and subsequently decreasing, in the coverage regime characteristic of the β_2 adstate ($\theta \leqslant 0.33$). The activation energies of desorption for the β_1 adstates ($\theta \geqslant 0.33$) decrease continuously with increasing coverage, whereas the preexponential factors remain essentially constant. The rate parameters for the Pt(110)-(1×2) surface are symmetric about a coverage equal to one-half of saturation of the β_2 adstate. This result has been interpreted as a manifestation of the occurrence of attractive adatom–adatom interactions between next-nearest-neighbors and repulsive interactions between nearest neighbors which, in this case, are apparently of approximately equal strength. The rate parameters for the Ir(110)-(1×2) surface appear to be asymmetric in the β_2 adstate coverage regime. However, these results should be treated with some caution, since a single high initial coverage ($\theta = 0.8$) was used to evaluate the entire coverage regime displayed in Fig. 16. It is probable that at the high heating rates (as large as 110 K s^{-1}) employed to evaluate the parameters, significant mixing occurred between the β_1 and β_2 adstates in the intermediate coverage regime ($0.2 \leqslant \theta \leqslant 0.45$). On the other hand, a mismatch in the strength of the attractive and repulsive interactions could contribute also to the observed asymmetry. In any case, the interpretation involving attractive adatom–adatom interactions at low coverages and repulsive interactions as the β_2 adstate approaches saturation remains valid also for the Ir(110)-(1×2) surface.

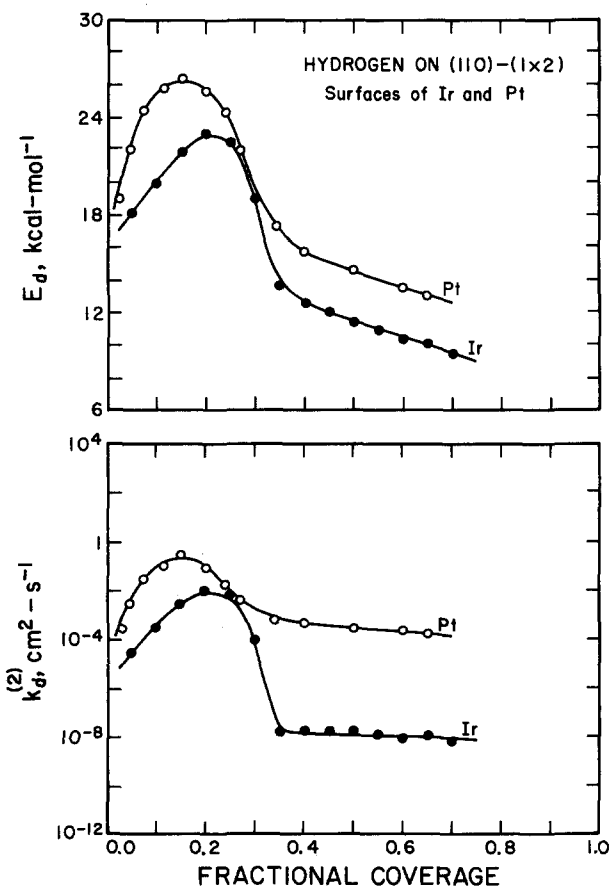


FIG. 16. Comparison of the desorption rate parameters, the activation energy (E_d), and the second-order preexponential factor ($k_d^{(2)}$), for hydrogen on the (110)-(1×2) surfaces of both iridium and platinum as a function of fractional coverage. The data for the Ir(110)-(1×2) surface are from Ibbotson *et al.* (Ref. 18).

V. CONCLUSIONS

The major conclusions of this work examining the chemisorption of hydrogen on both the Ir(111) and the Pt(110)-(1×2) surfaces can be summarized as follows:

(1) Hydrogen adsorbs dissociatively on the Ir(111) surface, the adsorption reaction described by second-order Langmuir kinetics with an initial probability of adsorption of 7×10^{-3}.

(2) The rate parameters describing the desorption of hydrogen from the Ir(111) surface are weakly dependent on coverage between fractional coverages of 0.1 and 0.3 and are given by $E_d \simeq 12.6$ kcal mol^{-1} and $k_d^{(2)} \simeq 2 \times 10^{-6}$ cm^2 s^{-1}. The activation energy of desorption decreases continuously with coverage above $\theta = 0.3$, indicative of the occurrence of repulsive interactions between nearest neighbor hydrogen adatoms.

(3) Both the variation in the desorption rate parameters at coverages below approximately 0.1 and the observation of an apparently higher probability of adsorption for this same coverage regime suggest that the Ir(111) surface examined

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here may contain a fraction of defect sites on the order of 2.5%–5.0%.

(4) Hydrogen adsorbs dissociatively on the Pt(110)-(1×2) surface into two distinct adstates which are designated as the β_2 and β_1. The ratio of the saturation densities of the two adstates, β_2:β_1, is 1:2. Adsorption into the higher binding energy β_2 adstate is described by first-order Langmuir kinetics with an initial probability of adsorption of 0.46. Adsorption into the β_1 adstate is described by second-order Langmuir kinetics with an "initial" probability of adsorption of 0.022.

(5) The rate parameters describing the desorption of hydrogen from the β_2 adstate on the Pt(110)-(1×2) surface depend strongly on the coverage. From the values for the zero coverage limit given by $E_d \simeq 18$ kcal mol^{-1} and $k_d^{(2)} \simeq 10^{-4}$ cm^2 s^{-1}, the rate parameters first increase to maximum values given by $E_d \simeq 26.5$ kcal mol^{-1} and $k_d^{(2)} \simeq 0.3$ cm^2 s^{-1} at $\theta = 0.15$, and subsequently decrease to approximately the values for the zero-coverage limit at $\theta = 0.32$, i.e., saturation of the β_2 adstate. The variation of the parameters in this coverage regime is symmetric about a coverage equal to one-half of saturation of the β_2 adstate, implicating the occurrence of attractive adatom–adatom interactions between next-nearest-neighbors and repulsive interactions between nearest neighbor hydrogen adatoms. In the coverage regime characteristic of the β_1 adstate, the activation energy decreases linearly with increasing coverage from $E_d \simeq 17$ kcal mol^{-1} at $\theta = 0.35$, whereas the preexponential factor maintains a constant value of $k_d^{(2)} \simeq 3\times10^{-4}$ cm^2 s^{-1}.

(6) Adsorption of hydrogen on the clean Pt(110)-(1×2) surface results in a continuous increase in the CPD, reaching a maximum value of 0.17 eV at a coverage of $\theta = 0.30$, i.e., nearly saturation of the β_2 adstate. Adsorption into the β_1 adstate results in a continuous decrease in the CPD, approaching a value of -0.50 eV at saturation. The initial increase in the CPD is attributed to the β_2 adstate being located below the image dipole plane of the surface, in the pseudo-fourfold hollow trough sites.

(7) The pre- or postadsorption of CO on the Pt(110)-(1×2) surface poisons hydrogen adsorption into the β_2 adstate, shifting preferentially its occupation to the β_1 adstates.

(8) Surface structure has been found to have a profound influence on the kinetics of adsorption of hydrogen. Comparisons of the results obtained on the (111) and (110)-(1×2) surfaces of both iridium and platinum suggest strongly that the *local* surface structure is the major determining factor. This viewpoint is supported by the striking similarities observed between the adsorption kinetics for the (111) surfaces and for the β_1 adstates on the (110)-(1×2) surfaces of both iridium and platinum, the latter adstates attributed to adsorption on the (111) microfacets of these reconstructed surfaces.

(9) The desorption kinetics have also been found to be influenced profoundly by the surface structure. The (110)-(1×2) surfaces of both iridium and platinum possess unique [with respect to the (111) surfaces] psuedo-fourfold hollow adsites that are associated with the higher binding energy β_2 adstates which are not observed on the (111) surfaces. Whereas both attractive and repulsive adatom-adatom interactions within the β_2 adstates are mediated by the (110)-(1×2) surfaces, only repulsive interaction are apparent on the (111) surfaces and for the β_1 adstates on the (110)-(1×2) surfaces.

ACKNOWLEDGMENT

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[60]An *approximate* heat of adsorption of D_2 on the Pt(111) surface was measured by Norton *et al.* (Ref. 22). The values that were reported (and used here) were approximate since a fixed gas temperature was employed. Nonetheless, the effects of a fixed gas temperature and the probable presence of an activation barrier to dissociative adsorption (on the order of 1 kcal mol^{-1}) are within other experimental uncertainties such that we may equate the heat of adsorption and the activation energy of desorption.

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